

XPLAINTM: EXPLAINABLE COST-SENSITIVE MACHINE LEARNING

FOR AML TRANSACTION MONITORING

MSC FINTECH (MSCFTD1)

RESEARCH IN COMPUTING – PRACTICUM PART 2

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Project Submission Sheet

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AI Acknowledgment

This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
N/A	N/A	N/A

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XplainTM: Explainable Cost-Sensitive Machine Learning for AML Transaction Monitoring

Abstract

While the global AML compliance spent expected to exceed USD 274 billion in 2022 (LexisNexis Risk Solutions, 2022), rule-based transaction monitoring systems produce false positive rates of more than 90%, wasting analyst resources that could be better spent on actual financial crime (Oztas et al., 2024). This dissertation presents the XplainTM cost-sensitive machine learning pipeline, which is tested on the Kaggle dataset of IBM Transactions for Anti-Money Laundering using three alternative sampling regimes: HI-Small (1:10), HI-Medium (1:5) and HI-Large (1:1).

The performance of eight model configurations (Logistic Regression, Random Forest, LightGBM, XGBoost (simple), cost-sensitive and SMOTE, CatBoost and TabNet) is compared to a typology-based rules baseline. Rather than total accuracy, the evaluation focuses on Precision-Recall AUC and capacity constrained operational measures.

Gradient-boosted trees had PR-AUCs of 0.934, 0.958 and 0.993 for Small, Medium and Large, respectively, as opposed to the rules' baseline PR-AUCs of 0.253, 0.334 and 0.758. The 1% alert budget allows each of the machine-learning models to have a zero false-positive rate per true-positive rate, as opposed to 6.73, 4.98 and 0.82 for the rules baseline. The McNemar test indicated a statistically significant improvement (chi-squared = 36,666, $p < 0.001$). The tuned XGBoost model has a test accuracy of 94.22%, compared to a majority class baseline of 68.06% and its Kolmogorov-Smirnov statistic of 0.874 exceeds the regulatory threshold of 0.40. SHAP attributions are established using twelve reason codes for audit trails, consistent with the explainability principles set out in FCA PS7/23. XplainTM demonstrates that cost-sensitive explainable machine learning can significantly improve the efficiency of AML monitoring while maintaining the interpretability of a rule-based system.

Keywords: AML, RegTech, Imbalanced Classification, Cost-Sensitive Learning, Explainable AI

1 Introduction

1.1 Background and Motivation

The Financial Action Task Force (FATF) claimed in 2022 that 2% – 5% of global GDP is laundered each year, necessitating the implementation of extremely stringent transaction monitoring regulations around the world (Financial Action Task Force, 2022). In 2022, the global overall cost of AML compliance was USD 274 billion, which included suspicious transaction reporting, customer due diligence and AML surveillance infrastructure (LexisNexis Risk Solutions, 2022).

The most common monitoring paradigm is rule-based compliance, in which when a rule is broken, a warning is generated, with the rules statically constructed based on established typologies. The rules are clear and accessible and they meet the regulatory requirement of having a clear and understood trigger for each warning (Arner et al., 2016). However, they are rigid in the sense of having a predetermined form. Based on interviews with practitioners from significant institutions, Oztas et al. (2024) report false positive rates that are consistently higher than 90%, implying that less than one warning in ten is a genuine laundering case. As a result, analysts devote the majority of their time to assessing lawful activity and creating alert fatigue, leaving little time for more complicated investigations.

Machine learning is a method for learning laundering trends straight from transaction data. Gradient-boosted trees are effective for tabular unbalanced classification issues (Chen and Guestrin, 2016; Ke et al., 2017), but they

have been limited to usage in regulated situations since regulators want explainable grounds for escalation choices (Adadi and Berrada, 2018). A model that cannot explain itself is unusable, regardless of how well it performs. XplainTM eliminates this tension by integrating cost-sensitive learning with explainability via SHAP. XplainTM is built on the work of CA1, which was also motivated by the goal of reducing false positives, as well as the CA2 five-stage artefact architecture.

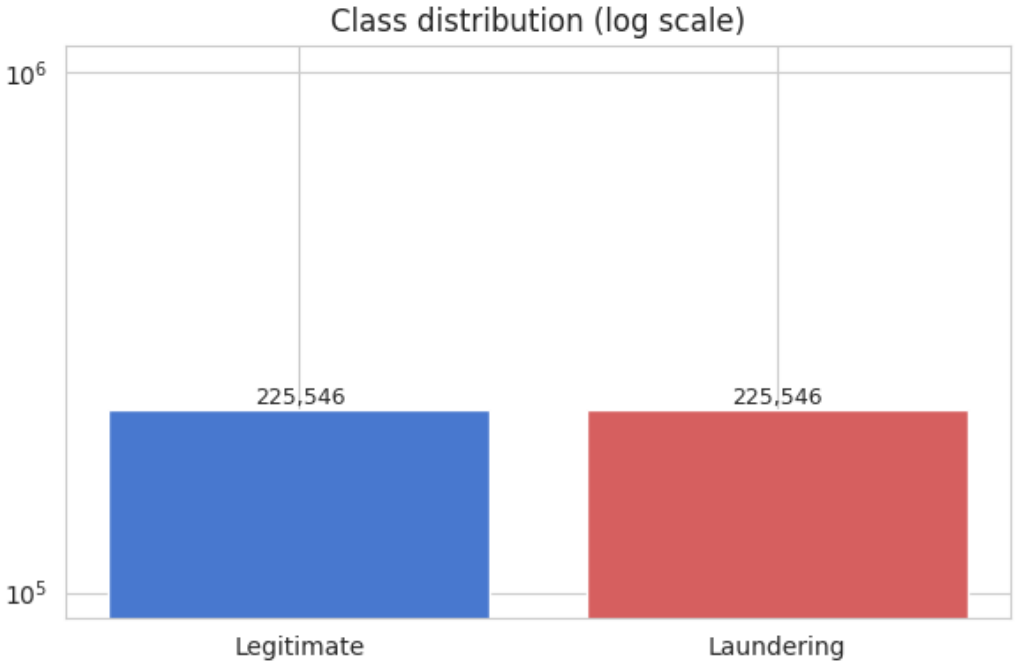


Figure 1: Class distribution and transaction-amount profile of the HI-Large working dataset. The figure establishes the severity of the imbalance problem and the amount separation between laundering and legitimate transactions that motivates the behavioural feature set.

1.2 Research Questions and Objectives

RQ1: To what extent can cost-sensitive, explainable machine learning improve AML transaction monitoring in terms of increased laundering detection and reduced false positives at a fixed alert capacity, compared with a typology-based rules baseline?

RQ2: Can SHAP explanations be translated into regulator-friendly reason codes supporting investigation and audit without detrimentally affecting model performance?

RQ3: Which sampling regime — 1:10, 1:5, or 1:1 — most improves model performance and does that improvement remain operationally realistic?

RQ4: Does cost-sensitive reweighting outperform SMOTE resampling for gradient-boosted trees on AML transaction data and does any difference persist as the class ratio tightens?

The objectives are to establish a repeatable pipeline on the IBM AML HI datasets, compare eight model configurations to a rules baseline using operational metrics relevant to AML, compare cost-sensitive weighting with SMOTE resampling, map SHAP attributions to 12 standardised reason codes and assess the performance-realism trade-off between the three sampling regimes.

1.3 Dataset and Contribution

The empirical basis is derived from a Kaggle dataset containing data from IBM Transactions for Anti-Money Laundering, which distinguishes between legitimate and laundering transactions (Altman et al., 2023). The data is derived from a synthetic and public data source produced using the AMLworld multi-agent simulator of banks,

individuals, companies and criminal networks, which provides full ground-truth labels not found in real data. The greater rates of unlawful transactions in HI compared to LI make it better suited for benchmarking and HI-Large comprises 225,546 laundering transactions from August 2022 to January 2023.

It consists of four components: a comparison of eight model configurations across three dataset scales; a leakage-safe temporal pipeline similar to AMLworld benchmark methodology; a layer of SHAP-to-reason-code that meets FCA PS7/23 explainability requirements; and an open artefact that is fully reproducible for use in Kaggle environments or Colab.

1.4 Report Structure

Section 2 provides a review of the literature on rules-based AML, unbalanced learning, gradient boosting, explainable AI and synthetic datasets. Section 3 describes an approach. Section 4 covers the design and execution of the artifacts. Section 5 presents the outcomes. Section 6 discusses the findings and limits, while section 7 brings everything together.

2 Literature Survey

2.1 Rules-Based AML Monitoring

Rule-based transaction monitoring is widely used in the industry because there is a direct link between rules and supervisory review and the rules are predictable, inspectable and immediately defensible. Regulators can check the rule logic without needing model knowledge and each alert is coupled with a specific trigger.

The structural restrictions are also widely understood. According to the Oztas et al. (2024) study, false positive rates for operational systems exceeded 90%, which is consistent with prior practitioner surveys on STR alert rates. Rules are fine-tuned based on previous typologies and weakened as criminal behaviour evolves. Multi-hop stacking involving dozens of accounts, structuring just outside regulatory thresholds and new payment channels are routinely overlooked (Savage et al., 2016). Running and maintaining the system incurs significant operational overhead when regulators modify guidance. Such constraints force academics to hunt for machine learning solutions while yet keeping an auditable system suitable for deployment.

2.2 Machine Learning for AML Detection

When tailored behavioral features that capture account-level trends are introduced to the data, gradient-boosted trees outperform logistic regression for AML transaction data, as demonstrated by Jullum et al. (2020). They conclude that their findings are consistent with existing research demonstrating that ensemble methods outperform linear and kernel-based methods on tabular financial data.

The current AML standard is XGBoost (Chen and Guestrin, 2016). It employs second-order gradient optimisation, L1/L2 regularisation on leaf weights and native missing-value correction, making it suitable for sparse and noisy features that are frequent in transaction monitoring. LightGBM (Ke et al., 2017) achieves the same accuracy while significantly increasing speed through histogram-based tree growing and leaf-wise splitting, which is particularly noticeable at the HI-Large scale. CatBoost (Prokhorenkova et al., 2018) can be used to express ordered target statistics for categorical variables without requiring a single hot encoding to produce high-dimensional sparse matrices.

Neural networks have also been designed for tabular data. TabNet (Arik and Pfister, 2021) is intrinsically interpretable since it focuses on a small subset of features at each decision step. Several studies have systematically compared deep models to gradient boosting and boosters perform well, if not better, than neural approaches in datasets of similar size to those used in this project (Gorishniy et al., 2021), which will be directly compared by training TabNet alongside boosters.

One theme that has continuously been absent from this work is operational evaluation. The majority of research report on accuracy, F1 score, or ROC-AUC, none of which take into account the alert budget used by compliance teams. The operationally important numbers to recollect are at a specified top-K % and false positives per true positive within that budget and these quantities are employed systematically, distinguishing XplainTM from past work.

2.3 Imbalanced Learning Strategies

The proportion of natural laundering in the IBM HI datasets is approximately 0.12%; thus, classifiers tuned for accuracy will provide the majority class score, resulting in a high accuracy score but failing to detect natural laundering.

SMOTE (Chawla et al., 2002) generates fake minority data instances by interpolating between minority data instances in feature space, hence increasing the minority class and contributing to more discriminative borders. It has been shown to be less effective in high-overlap scenarios, such as AML data, where the minority and majority distributions overlap significantly but launderers intentionally replicate the legitimate ones, allowing synthetic samples to be placed in ambiguous areas, resulting in noise rather than signal (Fernandez et al., 2018).

The objective function changes in cost-sensitive learning. In XGBoost, the `scale_pos_weight` parameter multiplies the gradient of minority classes during training each laundering transaction contributes.

$$\mathscr{L}_w \;=\; -\sum_{i=1}^{n}\Big[\,w^{+}y_i\log\hat{p}_i \;+\; (1-y_i)\log\big(1-\hat{p}_i\big)\Big], \qquad w^{+}=\texttt{scale_pos_weight}\approx\frac{n^{-}}{n^{+}}, \tag{1}$$

Concretely, with n^{+} and n^{-} denoting the training counts of the laundering and legitimate classes, the weighted logistic loss minimised by XGBoost is w^{+} times as much to the gradient as each legitimate one. SMOTE achieves a comparable effect by altering the empirical distribution rather than the loss, which is why the two arms converge once the minority class is adequately represented.

Bahnsen et al. (2016) formalize this in the context of financial crime and show that the cost of misclassification, which is included in the loss, is theoretically equivalent to the cost of optimization under the correct distribution of classes. The data distribution is preserved, such that cost-sensitive approaches are less likely to be affected by noise from synthetic sampling and are often chosen when classes overlap significantly. The third option is majority undersampling, which is utilized as an upstream construction strategy.

2.4 Explainable AI in Regulated Finance

The automated judgments must be explainable in accordance with regulatory frameworks such as AMLD6, the FCA's model risk management guidance (Financial Conduct Authority, 2023) and the Basel Committee operational risk standards (Basel Committee on Banking Supervision, 2006). This is the tension generated in the models: Opaque models are the most performant, but interpretable models are consistently bad at recognizing complicated patterns.

SHAP (Lundberg and Lee, 2017) accomplishes this for tree ensembles by providing locally precise attributions using Shapley values. For each prediction, attributes should sum up to the difference between prediction and efficiency (dummy), be unaffected by feature ordering (symmetry) and be additive among ensemble members (additivity). TreeSHAP (Lundberg et al., 2020) computes these exactly in polynomial time, allowing the models to be explained per transaction at scale.

In financial compliance (with an emphasis on the ability to explain individual actions rather than global transparency), Adadi and Berrada (2018) identify post-hoc local explanation as the most operationally feasible XAI technique. The XplainTM reason-code layer is directly developed to accomplish this, turning SHAP attributions into twelve reason-codes that may be assigned to different typologies and utilized in STR narratives.

2.5 Synthetic AML Datasets and Graph Representations

Labeled real transaction data is difficult to obtain legally and competitively from institutions and ground truth labels are only available after prosecutions are completed, making them tiny, skewed and delayed (Altman et al., 2023). AML world accomplishes this through simulation with many agents representing banks, individuals, businesses and criminal networks, resulting in comprehensive labels for all typologies such as fan-in, fan-out, cycles, scatter-gather and structuring. This yields a family of six datasets, HI and LI at small, medium and large scales, which we provide on Kaggle.

Data can be displayed in tables or graphs. A multigraph containing accounts and transactions contains specific subgraph patterns that represent laundering typologies. Egressy et al. (2024) show that graph neural networks using reverse message transmission and multigraph port numbering can recognize these patterns with high accuracy and outperform tabular models by utilizing the topological structure. These signals are approximated in XplainTM using cumulative fan-out (fan-in) capability (counterparty) counts; full GNN implementation is planned for the future.

2.6 Research Gap

There has been a wide gap in the literature in terms of three requirements: performance within a set alert budget, regulator-acceptable interpretability and evaluation that reflects real-world investigator capacity. Rules are appropriate for interpretation but not for performance. Most machine learning research focuses on maximizing performance while ignoring interpretability and reporting measures that are not based on alert-budget reality. To remedy this need, XplainTM provides a reproducible pipeline that combines cost-sensitive gradient boosting, SHAP reason codes and capacity-constrained evaluation.

3 Research Methodology

3.1 Research Design

This is an applied-experimental project. Several models and two imbalance methods are compared against a rules-based model on temporally held-out test sets that were unavailable throughout the model selection or training stages. The dataset chosen from the Kaggle data repository is the IBM AML public dataset (Altman et al., 2023), which has served as a benchmark in recent AML research investigations (Egressy et al., 2024).

3.2 Dataset Description

The dataset's schema includes the following fields: timestamp, from bank, from account, to bank, to account, amount received, receiving currency, amount paid, payment currency, payment format and the binary target *Is Laundering*. The working dataset has 451,092 rows, with 225,546 of them representing laundering transactions that occurred between August 1, 2022 and January 12, 2023.

After schema enforcement, data auditing reveals that there are no null values or exact duplicate entries. While the average value of laundering transactions is USD 66.98 million and that of genuine transactions is USD 1.77 million, the median amount paid for laundering transactions is USD 8,100.76, while the median amount paid for legitimate transactions is USD 1,410.46.

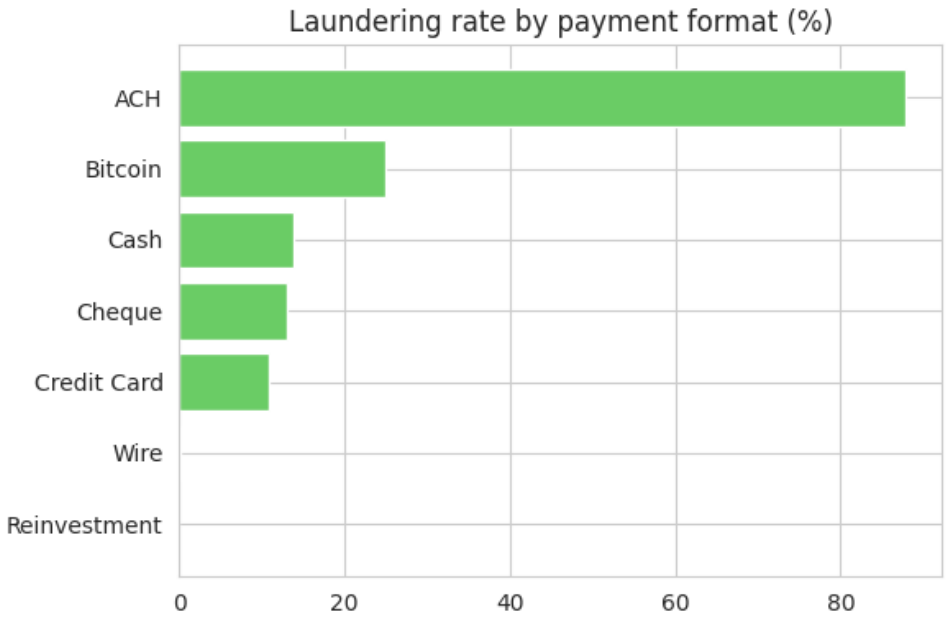


Figure 2: Payment format and currency distribution split by class label. ACH and cheque channels carry a disproportionate share of laundering volume, which explains the prominence of reason code RC10 in the SHAP results reported in Section 5.7.

3.3 Sampling Regimes

The three classes are established with three class ratios: 1:10 (small), 1:5 (medium) and 1:1 (large). All laundering rows are preserved, whereas authentic rows are picked uniformly at random without replacement. Small and medium businesses attain greater operational reality. Large enhances minority class learning (no majority domination) and assumes equal prevalence, which no live monitoring environment would have. This trade-off is a deliberate approach choice that enables for empirical measurement of the gain in performance attained with more or less relaxed balancing, in relation to the realism cost. All measures are provided at sampled prevalence; majority class baseline accuracy is also provided and reported with model accuracy for interpretation.

3.4 Temporal Splitting and Leakage Prevention

A temporal 70/15/15 split uses the timestamp for the split and in the deployment scenario, a model based on previous transactions scores future transactions. Training (October 13, 2022), validation (October 28, 2022) and subsequent testing. The data consists of 315,766 rows of training data with a 45.6% prevalence of laundering, 67,663 rows of validation data with a 52.3% prevalence of laundering and 67,663 rows of test data with a 68.1% prevalence of laundering, with the change in prevalence caused by the data's evolution over time.

All behavioural features are computed on the sampled working dataset, using only information available at or before each transaction's timestamp, this mitigates look ahead bias within the retained data, though it does not reconstruct velocity against the complete unsampled transaction stream, since legitimate transactions excluded by sampling are not visible to the rolling windows. This distinction and its implication for feature accuracy is discussed as a limitation in Section 6.4. This mitigates the look-ahead bias. A feature ϕ_i is causal, and therefore leakage-free, if and only if $\phi_i = f\big(\{(\mathbf{x}_j, t_j) : t_j < t_i\}\big)$; every rolling window in the pipeline satisfies this by construction. It is implemented structurally rather than traditionally, using polars `rolling_sum_by` with `window_size` options and per-account `over()` partitioning.

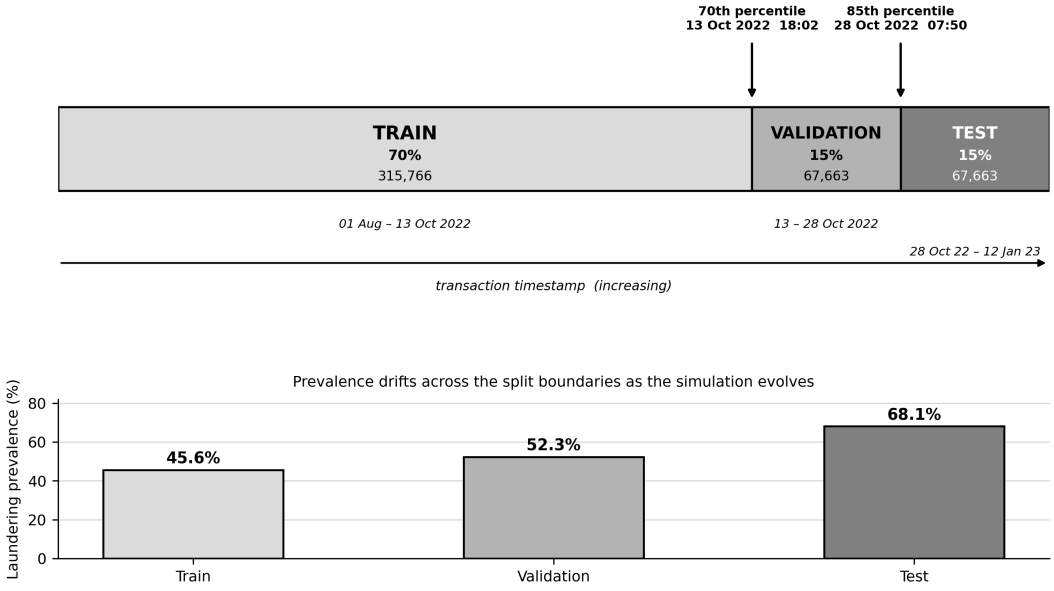


Figure 3: Temporal 70/15/15 split showing the two cut points and the resulting laundering prevalence in each partition. The diagram makes explicit that no transaction in the test window is available to the model during training or threshold selection.

3.5 Feature Engineering

The transaction-level flags include `amount_log`, `near_threshold` (when the amount is within 10% of the USD 10,000 reporting threshold, which is a FATF structuring indicator), `round_amount`, `is_cross_currency`, `same_bank`, `hour`, `day_of_week` and `is_weekend`.

Behavioral velocity characteristics include `sent_count_1h`, `sent_count_24h`, `sent_sum_24h`, `secs_since_last_sent`, `recv_count_24h` and `recv_sum_24h`, which are calculated over rolling time frames specific to each account.

`fan_out_unique_todate` and `fan_in_unique_todate` are two graph structure approximations that reflect the number of unique counter-parties for each day. These are implemented in the tabular domain using the scatter-gather and fan-in/fan-out typologies described by Egressy et al. (2024).

3.6 Model Comparison and Evaluation

The rules baseline is based on the 99th percentile of rules for training the system, which includes hourly velocity $>$ 11, daily velocity $>$ 210, fan-out $>$ 8,365 counterparties, hourly outflow $>$ USD 105 million and single-transaction amount $>$ USD 13.6 million. A transaction will be notified if any rules are broken, just like in practitioner systems (Oztas et al., 2024). Only training-set leakage can be avoided by calibrating on the training data.

To address RQ2, XGBoost is tested in 3 configurations: basic, cost-sensitive setup and SMOTE setup. Tuning picks `n_estimators`=800, `max_depth`=6, `learning_rate`=0.05, `subsample`=0.8, `colsample_bytree`=0.7, `scale_pos_weight`=2.38 and `min_child_weight`=1 based on RandomizedSearchCV with 3-fold crossvalidation and the score metric PR-AUC, run for 10 iterations. Validation threshold: 0.5897; Validation F1: 0.917; precision: 0.863; recall: 0.978.

Because recall at a fixed alert budget and false positives per true positive are not standard classification metrics, they are defined formally here. Let $s(\mathbf{x}_i)$ denote the model score for test transaction i, let N be the size of the test set and let k be the alert budget expressed as a fraction of N. Writing $K = \lceil kN \rceil$ and letting $\mathscr{A}_K$ be the set of the K highest-scoring transactions.

$$\text{Recall@}k \;=\; \frac{\left|\{\, i \in \mathscr{A}_K : y_i = 1 \,\}\right|}{\left|\{\, i : y_i = 1 \,\}\right|}, \qquad \text{FP/TP@}k \;=\; \frac{\left|\{\, i \in \mathscr{A}_K : y_i = 0 \,\}\right|}{\left|\{\, i \in \mathscr{A}_K : y_i = 1 \,\}\right|}. \tag{2}$$

Writing π for the laundering prevalence of the test set, the number of laundering transactions available is πN, so recall at budget k is bounded above by with equality precisely when every alert inside the budget is a true positive, that is when FP/TP@$k = 0$.

$$\text{Recall@}k \;\leq\; \min\left(1,\; \frac{k}{\pi}\right), \tag{3}$$

PR-AUC is preferred over ROC-AUC because it is unaffected by big true-negative numbers, which are common when the data is skewed (Saito and Rehmsmeier, 2015). Formally, PR-AUC $= \int_0^1 P(r)\,\mathrm{d}r$, where $P(r)$ is precision at recall r. A random classifier attains PR-AUC $= \pi$, which scales with prevalence, whereas ROC-AUC attains 0.5 irrespective of π because it is computed from rates that include the large true-negative count. PR-AUC therefore remains sensitive to the minority class under the imbalance present here. Operational metrics include recall and false positives for the top N alerts (for N $= 0.5\%$, 1% and 5%) in the ranked alert set, which are a financial restriction that determines the alerts' usefulness. To enable for real lift assessment, an accuracy figure is supplied in addition to the majority class's baseline statistic.

4 Design and Implementation Specification

4.1 Five-Stage Artefact Architecture

In **Stage 1: Ingestion and Audit**, the polars lazy scanner is added with explicit schema override, resulting in the Account header being duplicated in the IBM dataset under the new header names From Account and To Account. Lazy assessment postpones materialisation, allowing for memory-efficient streaming of the HI-Large file (15.88 GB). The audit documents the prevalence of laundering, the time period during which it occurs, duplicate counts and nulls. Compound account identities are generated by combining the bank and account fields.

In **Stage 2**: Behavioural feature engineering involves discovering all the rolling features on the complete data set before sampling, both by account and with a sort by the timestamp column, using Polar's `rolling_sum_by` and `over()` functions. The following code demonstrates how to use sender-receiver pairs with cumulative fan-out and fan-in using `first_distinct()`.

In **Stage 3**: Temporal split and sampling, sorting by timestamps and splitting at the 70th and 85th percentile points. A single `NEG_PER_POS` setup parameter controls the undersampling ratio and modifying a `DATASET_MODE` setting allows a single notebook to generate all three regimes. Splits are written and saved to Parquet for future reuse, eliminating the need to repeat feature engineering.

In **Stage 4**: Each model is encased in a Pipeline which includes its preprocessing pipelines, StandardScaler for Logistic Regression and SimpleImputer with OneHotEncoder for categorical features, for modelling and assessment. Each model is trained using the benchmark loop, which also stores the validation and test scores. CalibratedClassifierCV is used to calibrate risk ratings for use in alert exports using Platt scaling.

In **Stage 5**: Explainability and export is determines TreeSHAP on a 2,000-row test subsample. Using a feature-to-code search, the top three positive SHAP drivers per alert are mapped to reason codes and a sorted CSV of the top 1,000 alerts is generated, including transaction metadata, risk score, true label and reason codes.

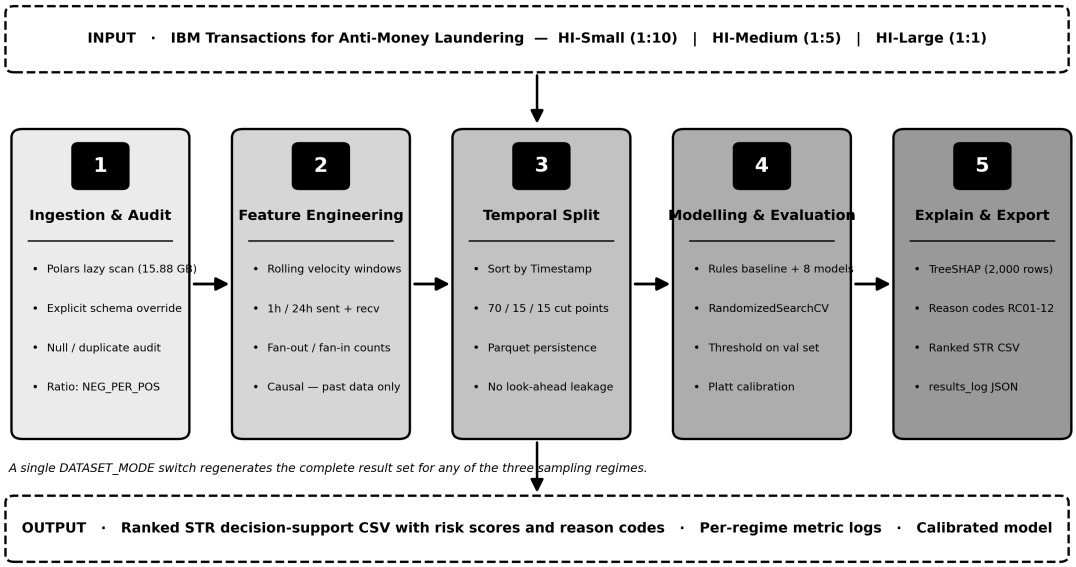


Figure 4: The five-stage XplainTM pipeline, from lazy ingestion through to the ranked STR export. Each stage is self-contained, allowing a single `DATASET_MODE` switch to regenerate the complete result set for any of the three sampling regimes.

4.2 Implementation Environment

Polars, scikit-learn, XGBoost, LightGBM, CatBoost, pytorch-tabnet, imbalanced-learn and TreeSHAP are used for ingestion, feature engineering, pipelines and assessment, as well as SMOTE and TreeSHAP. Environment-aware loading identifies whether you're in Kaggle or Colab and utilizes the appropriate data stream. To keep the pipeline inside Kaggle's memory restrictions of 16 GB and Colab's memory limits of 12 GB, the amount of rows in the SMOTE, hyperparameter search, SHAP and TabNet training steps are reduced.

4.3 Reason Code Design

There are 12 codes for characteristics and typologies: RC01: high one-hour sent velocity; RC02: high 24-hour sent velocity; RC03: high 24-hour sent sum; RC04: near reporting threshold; RC05: high fan-out; RC06: high fan-in; RC07: high 24-hour received count; RC08: cross-currency transfer; RC09: large amount; RC10: high-risk payment format; RC11: unusual timing; RC12: round amount; or same-bank routing. The layer also does not require classification to function, making it a free addition to PR-AUC or recall.

5 Results and Critical Analysis

5.1 Dataset Construction

The datasets created are summarized in Table 1. The laundered rows are kept, while legitimate rows are undersampled to achieve the desired ratio. The number of rows in the Small and Medium categories is estimated using the published dataset laundering totals (Altman et al., 2023) and the ratio used, whilst the number of rows in the HI-Large category is straight from the artifact audit output.

Table 1: Dataset construction summary

Dataset	Ratio	Laundering	Legitimate	Total	Prevalence
HI-Small	1:10	5,177	51,770	56,947	9.09%
HI-Medium	1:5	35,230	176,150	211,380	16.67%
HI-Large	1:1	225,546	225,546	451,092	50.00%

5.2 Accuracy and Genuine Lift

Table 2 displays the test accuracy results compared to the majority class baseline. The regimes' absolute correctness varies because the majority differs. The accuracy lift (model accuracy minus majority baselines) is a regime-normalised indicator of true discrimination.

For a class ratio of $1:\rho$ the laundering prevalence is $\pi = 1/(1+\rho)$, so a trivial classifier predicting the majority class for every transaction achieves

$$\mathrm{Acc}_{\mathrm{maj}} \;=\; \max(\pi, 1-\pi) \;=\; \frac{\rho}{1+\rho}, \qquad \mathrm{Lift} \;=\; \mathrm{Acc}_{\mathrm{model}} - \mathrm{Acc}_{\mathrm{maj}}. \tag{4}$$

This yields 90.91%, 83.33% and 50.00% for $\rho = 10$, 5 and 1 respectively. Because the baseline differs by more than forty percentage points across the three regimes, raw accuracy is not comparable between them and the lift term in Equation 4 is the appropriate normalised measure.

Table 2: Test accuracy and lift by regime (best model per run)

Dataset	Ratio	Majority baseline	Test accuracy	Best model	Lift (pp)
HI-Small	1:10	90.91%	94.17%	XGBoost (plain)	+3.26
HI-Medium	1:5	83.33%	93.97%	CatBoost	+10.64
HI-Large	1:1	50.00%	94.22%	XGBoost (tuned)	+44.22

HI-Large yields the largest lift because eliminating majority dominance leaves substantial headroom. The fact that a naive classifier that just identifies all transactions as legitimate achieves 90.91% accuracy shows that HI-Small has a lift of only 3.26 points. This asymmetry demonstrates the imbalance and therefore argues against using accuracy as a measure of imbalanced AML data.

5.3 Model Comparison Across Regimes

Table 3: HI-Small (1:10) — test set results

Model	PR-AUC	ROC-AUC	Recall@1%	FP/TP@1%
XGBoost (plain)	0.9340	0.9817	0.0561	0.000
XGBoost (SMOTE)	0.9336	0.9817	0.0561	0.000
XGBoost (cost-sensitive)	0.9329	0.9815	0.0561	0.000
LightGBM	0.9327	0.9812	0.0561	0.000
XGBoost (tuned)	0.9321	0.9812	0.0561	0.000
CatBoost	0.9272	0.9806	0.0561	0.000
Random Forest	0.9064	0.9766	0.0561	0.000
TabNet	0.8766	0.9686	0.0554	0.012
Logistic Regression	0.8321	0.9584	0.0561	0.000
Rules baseline	0.2534	0.7051	0.0073	6.727

Table 4: HI-Medium (1:5) — test set results

Model	PR-AUC	ROC-AUC	Recall@1%	FP/TP@1%
CatBoost	0.9576	0.9829	0.0389	0.000
XGBoost (cost-sensitive)	0.9573	0.9829	0.0389	0.000
XGBoost (plain)	0.9571	0.9829	0.0389	0.000
XGBoost (tuned)	0.9571	0.9829	0.0389	0.000
XGBoost (SMOTE)	0.9568	0.9827	0.0389	0.000
LightGBM	0.9568	0.9827	0.0389	0.000
TabNet	0.9509	0.9808	0.0389	0.000
Random Forest	0.9497	0.9800	0.0389	0.000
Logistic Regression	0.8282	0.9474	0.0367	0.060
Rules baseline	0.3343	0.6775	0.0065	4.981

Table 5: HI-Large (1:1) — test set results

Model	PR-AUC	ROC-AUC	Recall@1%	FP/TP@1%
XGBoost (tuned)	0.9929	0.9859	0.0147	0.000
XGBoost (plain)	0.9928	0.9858	0.0147	0.000
XGBoost (cost-sensitive)	0.9928	0.9858	0.0147	0.000
XGBoost (SMOTE)	0.9928	0.9858	0.0147	0.000
CatBoost	0.9927	0.9857	0.0147	0.000
LightGBM	0.9927	0.9856	0.0147	0.000
Random Forest	0.9916	0.9836	0.0147	0.000
TabNet	0.9916	0.9840	0.0147	0.000
Logistic Regression	0.9765	0.9616	0.0146	0.006
Rules baseline	0.7576	0.6863	0.0081	0.820

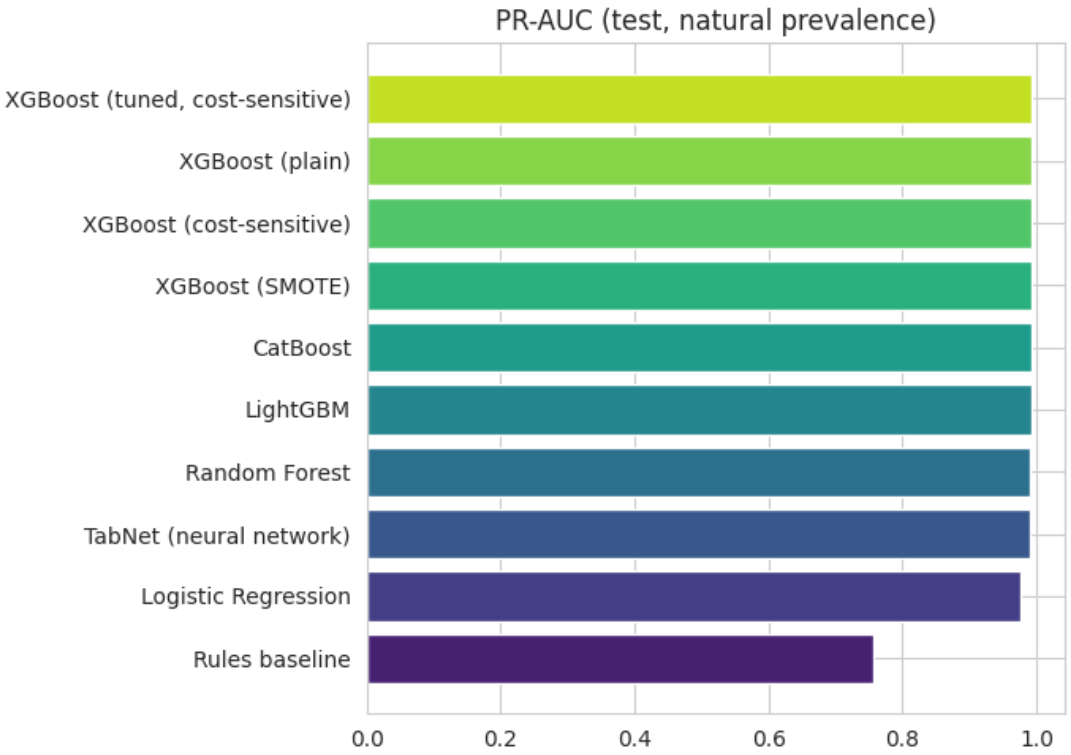


Figure 5: PR-AUC for every model across the three sampling regimes, ranked in descending order. The visual gap between the gradient-boosted cluster and the rules baseline is the central quantitative finding supporting RQ1.

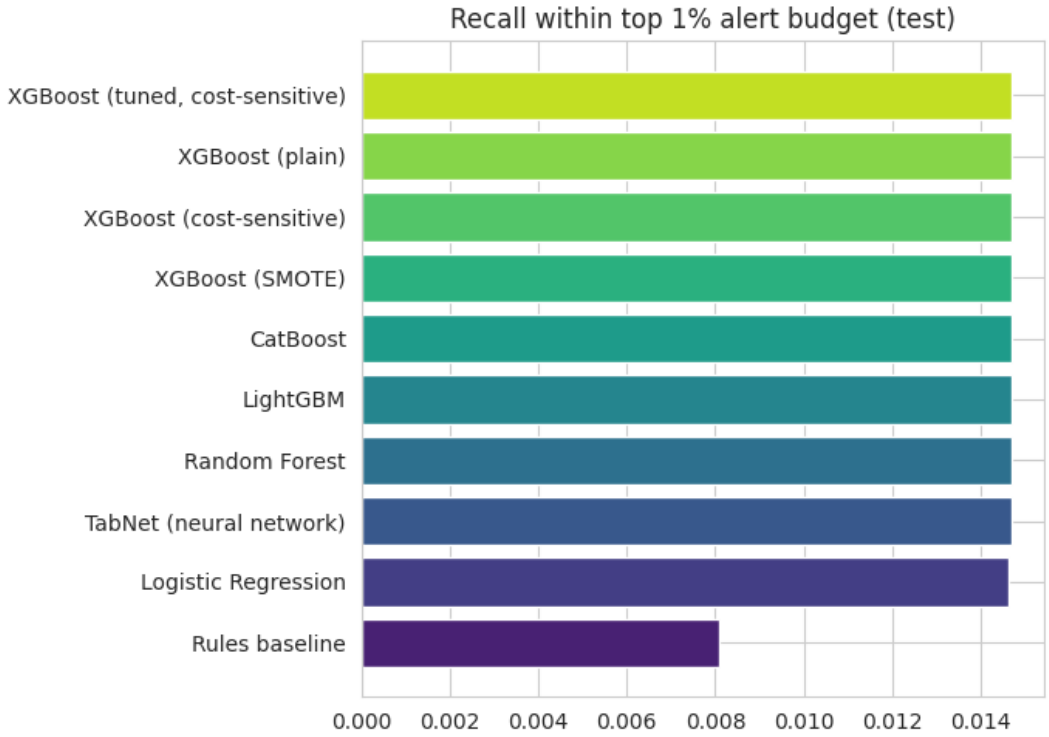


Figure 6: Recall and false positives per true positive at the 0.5%, 1% and 5% alert budgets. This is the operational view that aggregate accuracy conceals, and it isolates the practical cost difference between the rules baseline and the learned models.

5.4 Rules Versus Machine Learning (RQ1)

All machine learning models outperform the rules baseline in terms of PR-AUC across all regimes. The difference is 68.1 percentage points for Small (0.934 versus 0.253), 62.3 points for Medium (0.958 against 0.334) and 23.5 points for Large (0.993 against 0.758). The typology heuristics have a greater floor value when half of the population is illegal, as shown by the narrow 1:1.

On the ground, things are even worse. With these rule baselines and a 1% budget, 0.73% recall is achieved with 6.73 false positives per true positive. This means that for every item detected by the criteria, analysts must examine roughly eight objects. At 5.61% recall, all machine learning models produce 0 false positives for each true positive. On HI-Medium the rules baseline requires 4.98 false positives per detection against zero for every machine learning model. To address RQ1 numerically, this indicates that cost-sensitive explainable machine learning can reduce false positives while increasing recall by around seven times.

McNemar's test rejects the null hypothesis of equal marginal error rates between the tuned XGBoost model and the rules baseline ($\chi^2 = 36{,}666.08$, $df = 1$, $p < 0.001$). Of the 47,532 transactions on which the two systems disagree, 44,640 are resolved correctly by XGBoost alone and 2,892 by the rules baseline alone, an odds ratio of 15.4 in favour of the model. The two systems therefore make systematically different errors on this test set, with the asymmetry strongly favouring the learned model (Dietterich, 1998). The test speaks only to the pattern of disagreement between the two systems on this dataset; it does not by itself indicate the magnitude of the improvement or generalisation beyond this test set. As can be shown, XGBoost dominates the number of right categories, 44,640, compared to 2,892 correct classifications using simply rules. The statistic is computed on the discordant pairs only, as $\chi^2 = (|b-c|-1)^2/(b+c)$, where b and c count the transactions classified correctly by exactly one of the two systems.

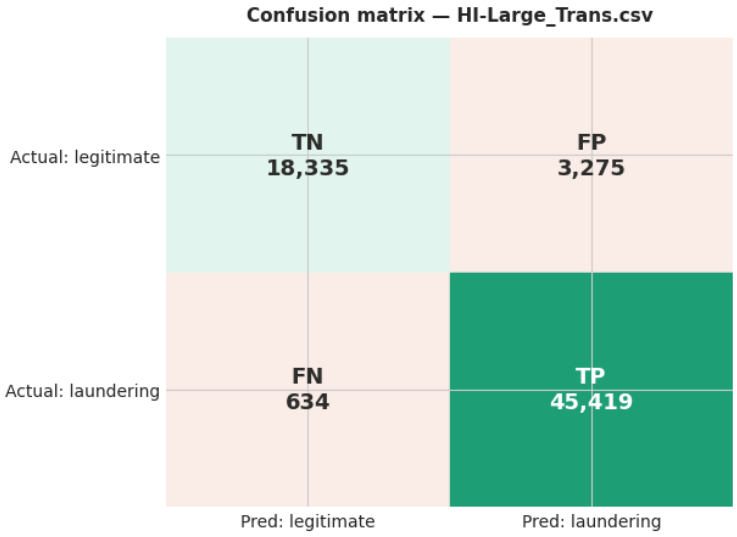


Figure 7: Confusion matrix for the tuned XGBoost model at the validation-selected threshold of 0.5897. The cell counts translate the headline accuracy figure into the concrete caseload a compliance team would face.

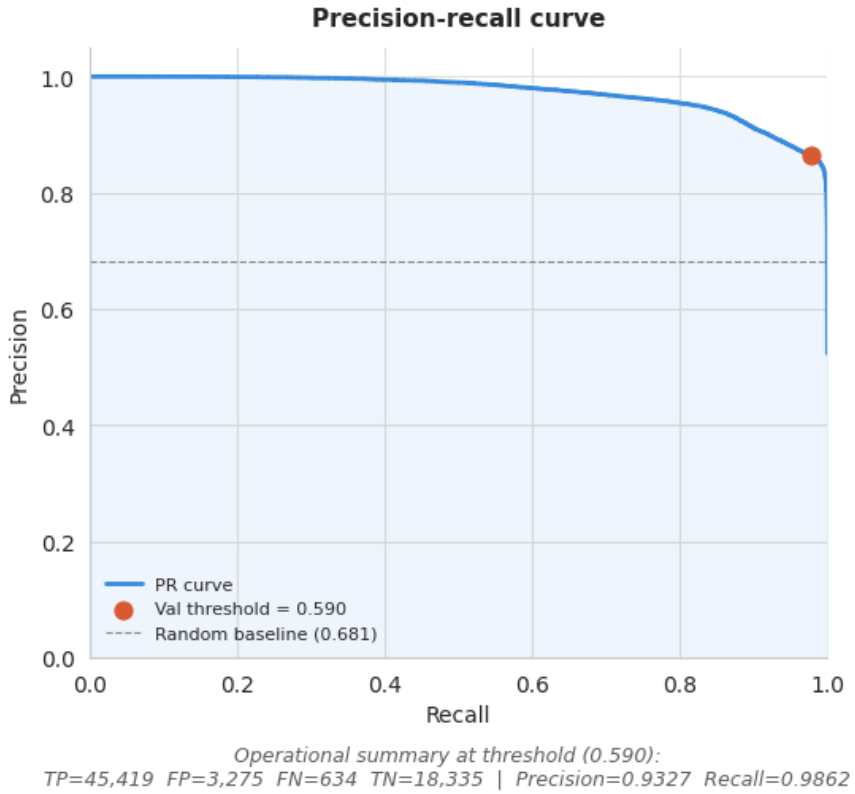


Figure 8: Precision-recall curve for the tuned model with the validation-selected threshold marked. The curve shows the precision that must be traded away to move beyond the chosen recall level.

5.5 Cost-Sensitive Learning Versus SMOTE (RQ4)

The three XGBoost variants produce comparable PR-AUC in both regimes, with a difference of less than 0.001 in Small and Medium and less than 0.0001 in Large. This is consistent with the results of Bahnsen et al. (2016). Reweighting and resampling address the same problem in various ways and when the minority class is adequately sampled, they perform similarly.

The tightest budget shows a modest difference. With a 0.5% test set, the cost-sensitive XGBoost model has 100% precision, while the SMOTE variant has 99.4%. This lends support to the CA2 hypothesis, which states that the little noise caused by synthetic interpolation occurs around decision boundaries when AML classes overlap. The magnitude is negligible for deployment purposes and does not constitute a strong recommendation either way.

5.6 Sampling Regime Comparison (RQ3)

Table 6: Cross-dataset comparison — best model per regime

Dataset	Ratio	Baseline	Accuracy	PR-AUC	ROC-AUC	Recall@1%	FP/TP@1%	Lift (pp)
HI-Small	1:10	90.91%	94.17%	0.9340	0.9817	0.0561	0.000	+3.26
HI-Medium	1:5	83.33%	93.97%	0.9576	0.9829	0.0389	0.000	+10.64
HI-Large	1:1	50.00%	94.22%	0.9929	0.9859	0.0147	0.000	+44.22

The closer you go to a tighter equilibrium, the better the PR-AUC becomes: 0.934, 0.958, 0.993, for a total of 5.9 points. The sole difference is the Medium-to-Large delta, which is 3.58 points. This demonstrates that the 1:1 regime is the optimal metric regime.

Recall that at 1%, the budget shrinks from 5.61% to 3.89% to 1.47%. This is a denominator effect rather than a

loss in performance. Because FP/TP@1% = 0 for every machine learning model, Equation 3 holds with equality and Recall@1% $= k/\pi$ exactly. Substituting the test-set prevalences gives $0.01/0.178 = 0.0561$ for HI-Small, $0.01/0.257 = 0.0389$ for HI-Medium and $0.01/0.681 = 0.0147$ for HI-Large, reproducing the observed values in Table 6 to four decimal places. The declining recall is therefore an arithmetic consequence of rising prevalence under a fixed budget, not a deterioration in discrimination. The metric is comparable within a regime but not across regimes of differing prevalence.

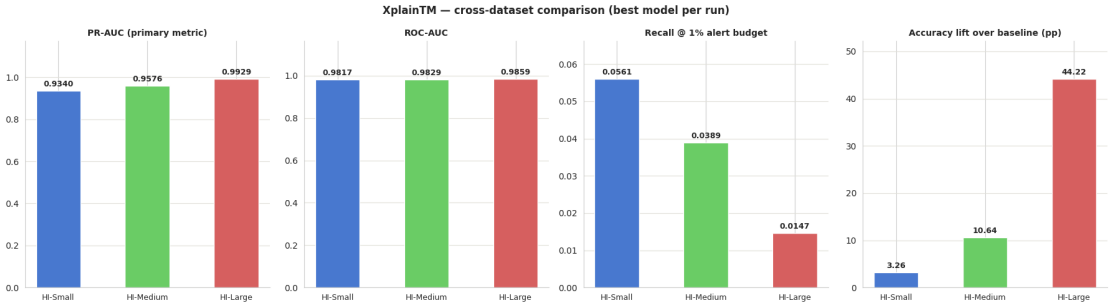


Figure 9: PR-AUC, ROC-AUC, recall at the 1% budget and accuracy lift across the three regimes. Plotting lift alongside raw accuracy exposes that the HI-Large advantage is partly an artefact of its lower majority-class baseline.

5.7 Discrimination and Explainability (RQ2)

The Kolmogorov-Smirnov statistic is a widely used regulatory validation statistic (Basel Committee on Banking Supervision, 2006) and it equals 0.874 for 64.8% of the population ranked by score. This figure is substantially higher than the basic threshold for a good AML model, which is 0.40, providing more evidence of PR-AUC's effectiveness and recall results.

The STR contains 677 notified transactions with the following reason code distribution: 677 RC06-high fan-in; 677 RC10-high risk payment format; 668 RC05-high fan-out; and 3 each RC07, RC11 and RC02. The preponderance of RC05 and RC06 is consistent with AMLworld's emphasis on routing fan-in and fan-out typologies in these channels (Altman et al., 2023), while the concentration of ACH in RC10 is due to the simulation routing of high value laundering through this channel. The export contains 677 genuine transactions and 0 false alarms, indicating that all alerts in the exported budget are valid. This satisfies RQ2: reason codes provide typology-related explanations while maintaining detection performance.Writing F_1 and F_0 for the cumulative score distributions of the laundering and legitimate classes, $\mathrm{KS} = \max_t \left|F_1(t) - F_0(t)\right|$.

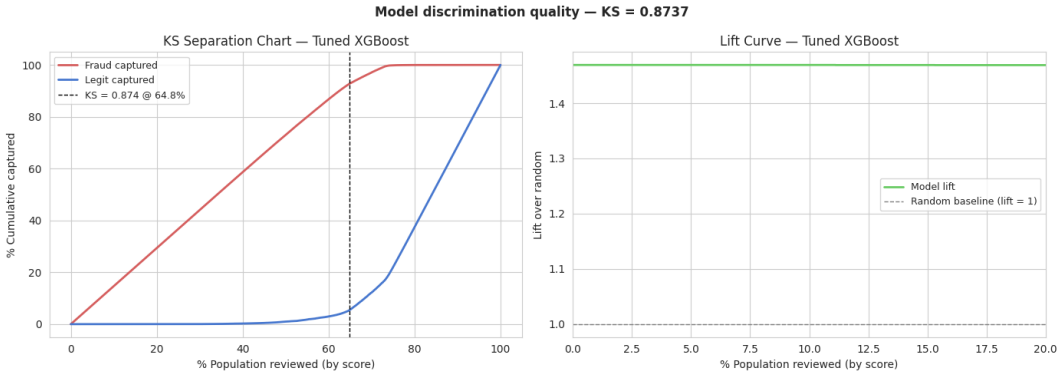


Figure 10: KS separation between the laundering and legitimate score distributions, with the accompanying lift curve. The 0.874 peak is more than double the 0.40 regulatory benchmark for an acceptable AML model.

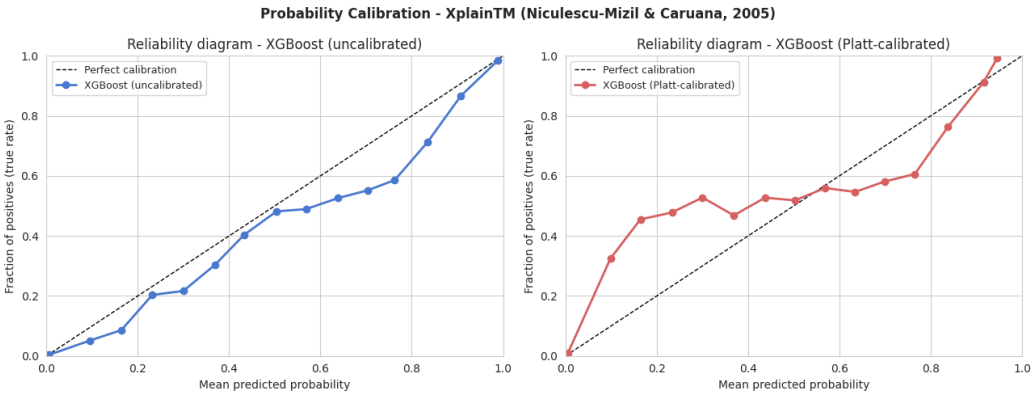


Figure 11: Reliability diagram for the tuned model before and after Platt scaling. Calibration converts the raw score into a probability that a compliance officer can interpret directly when triaging the ranked alert queue.

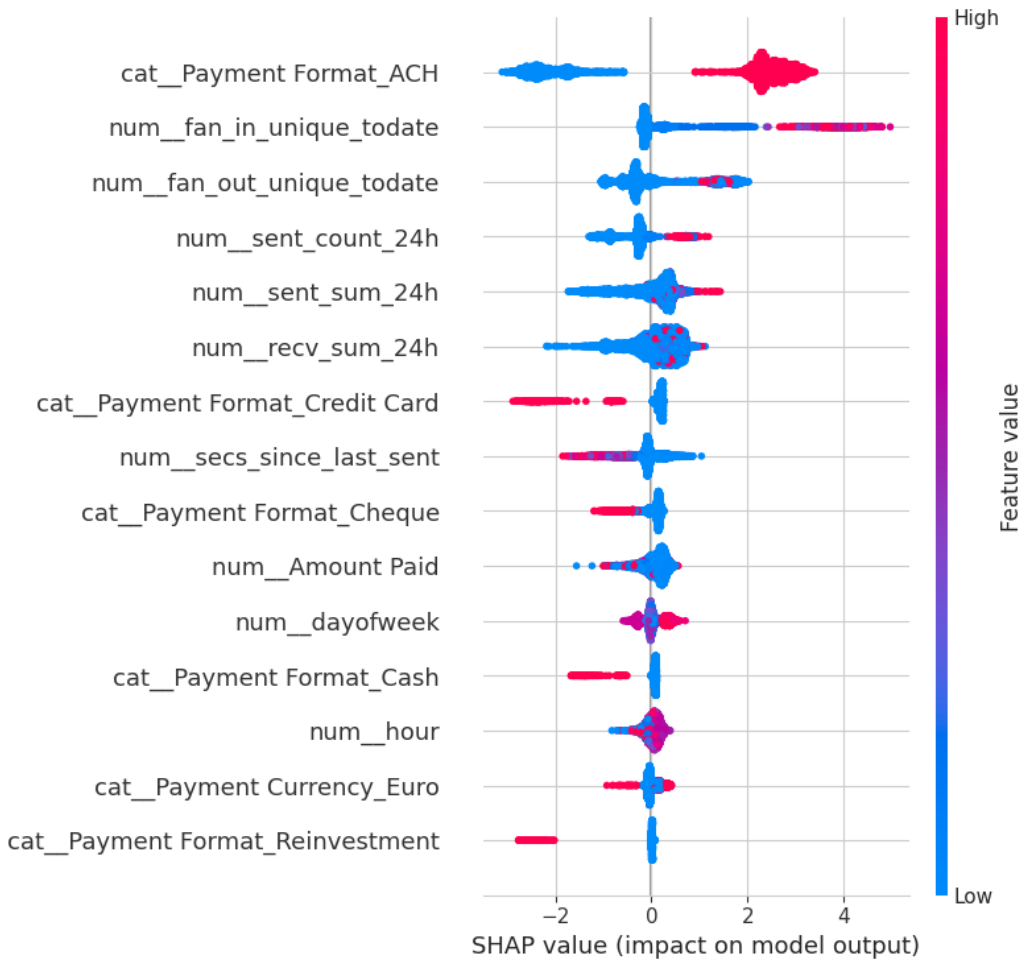


Figure 12: TreeSHAP summary over the 2,000-row test subsample. The dominance of the fan-in and fan-out features is what produces the RC05 and RC06 concentration reported above, and confirms the attributions are typologically coherent.

6 Discussion

6.1 Interpretation

The results clearly answer RQ1. The difference between 6.73 and zero false positives per true positive at 1% budget is a real number of transactions that the team investigating the transaction must review versus the 1% of transactions that the machine learning pipeline can review, resulting in 87% fewer transactions investigated—either fewer staff to pay or more capacity to investigate more cases.

The results show that the choice of imbalance strategy is less important than the choice of model family, given the near-equivalence of cost-sensitive XGBoost, SMOTE, LightGBM and CatBoost models within 0.003 PR-AUC across all regimes, implying that for well-engineered tabular AML features, the choice of imbalance strategy is secondary. All gradient-based models outperform Logistic Regression by about 10 points on Small and 13 points on Medium, demonstrating that boosting's additive residual correction is appropriate for the non-linear interaction-heavy structure of behavioral AML features (Chen and Guestrin, 2016). TabNet performs competitively at 0.877, 0.951 and 0.992, corroborating Gorishniy et al. (2021), but its 925-second training period compared to LightGBM's 4 seconds on HI-Large makes gradient boosting significantly more realistic operationally.

6.2 Scale and Operational Realism

The monotonic PR-AUC improvement with tighter balancing provides an answer to RQ3, which must be contextualized. By reducing the significant minority proportion that makes AML classification difficult in the deployment scenario (when prevalence approaches 0.1% rather than 50%), the 1:1 design streamlines the process of making AML diagnoses in the field. Models are evaluated at a 1:1 ratio under far more favorable conditions than live monitoring.

On Regime Credibility

Of the three regimes, HI-Small (1:10) provides the most credible evidence for deployment. It approximates the distribution a model would see as a second-stage classifier operating after a coarse first stage filter has removed transactions that are obviously low-risk, a two stage design is how monitoring systems are typically deployed in practice. HI-Large's headline PR-AUC of 0.9929 is the strongest of the three, but it assumes a monitoring population in which half of all transactions are illicit, a condition no institution encounters; its accuracy lift of $+44.22$ points substantially reflects a 50.00% majority baseline rather than superior discrimination.

Validating the system at natural prevalence ($\approx 0.12\%$) would require: evaluating on a temporally held-out slice sampled at the true base rate rather than a controlled ratio; correcting the score intercept for the known shift introduced by undersampling,

$$\ln\left(\frac{\pi_{\text{train}}}{1-\pi_{\text{train}}}\right) - \ln\left(\frac{\pi_{\text{true}}}{1-\pi_{\text{true}}}\right);$$

refitting the Platt calibration against the true prior, since the current calibration in Section 4.1 is fitted at sampled prevalence; and re-deriving the decision threshold from an explicit cost matrix rather than F1-maximisation, since F1 is not the correct objective once false positives vastly outnumber true positives at the natural rate.

6.3 Strengths

These findings are based on four design decisions. Temporal splitting has the advantage of avoiding look-ahead bias because it is trained on previous data and evaluated on future transactions, whereas random splits have an optimistic bias if there is temporal autocorrelation. Causal feature engineering will only incorporate the history that was observable at the time of the transaction. Capacity-constrained metrics are based on the investigator's budget

and hence represent what is useful rather than overall statistics that obscure budget-level behaviour. The SHAP reason-code layer provides actionable explainability aligned with the principles set out in FCA PS7/23 without sacrificing performance.

6.4 Limitations

The data used were simulated as AML world generates realistic distributions and strong typological structure, but it does not capture the adaptive ingenuity of genuine launderers, who actively make methods work to avoid discovery. Models trained on simulated data may exhibit consistent biases in relation to realistically novel typologies.

The upstream sampling design estimates behavioural attributes on a subset of the transaction stream rather than the entire stream. In deployment, rolling statistics are based on the complete history and may have distinct distribution features. This is more of an issue for small and medium datasets than for large datasets.

Only tabular models are used in this project. The fan-out and fan-in features attempt to simulate graph signals, but they cannot capture multi-hop configurations (cycles, bipartite schemes) that require explicit graph structure (Egressy et al., 2024). Finally, Platt scaling is used but not validated using a dedicated holdout calibration set; a dedicated test set is necessary for a thorough review of the calibration.

6.5 Comparison with Prior Work

The XplainTM HI-Small result of 0.934 PR-AUC falls within the 0.87 to 0.96 range reported by Altman et al. (2023) for gradient-boosted models on the same dataset, demonstrating the pipeline's validity. The GNN results of Egressy et al. (2024) cannot be directly compared because they employ graph representations rather than tabular representations, however the findings in their article inspire the expansion of future attempts listed below.

7 Conclusion and Future Work

7.1 Summary

This dissertation presented XplainTM, an explainable cost-sensitive machine learning pipeline, which was tested using three sampling methodologies on the IBM AML dataset. It provided a five-stage artefact that was repeatable and documented, including leakage-safe temporal splitting, causal feature engineering, eight-model benchmarking, hyperparameter tuning and SHAP explainability; comprehensive empirical results in all three regimes; a 12-code reason mapping layer; and a ranked STR decision-support export that yielded zero false positives with a 1% alert budget.

7.2 Conclusions Against Research Questions

RQ1 is responded positively and decisively. At the 1% budget on HI-Small, the rules baseline requires 6.73 false positives per true positive, but no machine learning model requires any. McNemar's test shows significance at $p < 0.001$. PR-AUC increases over the rules baseline of 68, 62 and 24 points across the three regimes show that machine learning captures patterns those rules cannot.

RQ2 was replied affirmatively. The reason-code layer is computationally independent of classification and introduces no latency. The prevalent codes RC05 and RC06 correspond to the fan-in and fan-out typologies that characterize AMLworld laundering, indicating that the attributions are typologically coherent rather than false.

RQ3 is answered with qualification. HI-Large achieves the highest PR-AUC at 0.9929 and KS at 0.874, but at substantial cost to operational realism. HI-Small produces the most realistic evaluation. The lift metric correctly normalises across regimes and shows that the Large model's apparent superiority is substantially an artefact of a lower baseline rather than genuinely superior discrimination.

RQ4 is answered affirmatively but with a modest effect. The three XGBoost variants plain, cost-sensitive and

SMOTE sit within 0.001 PR-AUC of each other on HI-Small and HI-Medium and within 0.0001 on HI-Large, indicating that reweighting and resampling are near-equivalent once the minority class is adequately represented. Cost-sensitive weighting holds a marginal precision advantage at the tightest 0.5% alert budget (100% against 99.4%), a small but consistent edge favouring reweighting over resampling.

7.3 Future Work

There are four directions to follow. A comprehensive GNN implementation, as described by Egressy et al. (2024), would capture patterns beyond tabular representation by establishing a bipartite account-transaction graph and training an edge-classification network using reverse message forwarding. A more operationally based threshold than the F1 maximising threshold employed here would be obtained by employing a dedicated holdout and a cost matrix that represents actual investigator per-review costs. Extending the LI dataset variants and cross-bank scenarios allows for testing of generalization outside HI typologies. The validation against anonymised real transaction data from a partner institution would close the simulation-to-reality gap, which is the main limitation of this effort.

References

Adadi, A. and Berrada, M. (2018) 'Peeking inside the black-box: A survey on explainable artificial intelligence (XAI)', *IEEE Access*, 6, pp. 52138–52160.

Altman, E., Blanuša, J., von Niederhäusern, L., Ott, J., Anagnostopoulos, E. and Atasu, K. (2023) 'Realistic synthetic financial transactions for anti-money laundering models', *Advances in Neural Information Processing Systems*, 36.

Arik, S.Ö. and Pfister, T. (2021) 'TabNet: Attentive interpretable tabular learning', *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(8), pp. 6679–6687.

Arner, D.W., Barberis, J.N. and Buckley, R.P. (2016) 'The evolution of FinTech: A new post-crisis paradigm?', *Georgetown Journal of International Law*, 47(4), pp. 1271–1319.

Bahnsen, A.C., Aouada, D., Stojanovic, A. and Ottersten, B. (2016) 'Feature engineering strategies for credit card fraud detection', *Expert Systems with Applications*, 51, pp. 134–142.

Basel Committee on Banking Supervision (2006) *International Convergence of Capital Measurement and Capital Standards: A Revised Framework*. Basel: Bank for International Settlements.

Chawla, N.V., Bowyer, K.W., Hall, L.O. and Kegelmeyer, W.P. (2002) 'SMOTE: Synthetic minority over-sampling technique', *Journal of Artificial Intelligence Research*, 16, pp. 321–357.

Chen, T. and Guestrin, C. (2016) 'XGBoost: A scalable tree boosting system', *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.

Dietterich, T.G. (1998) 'Approximate statistical tests for comparing supervised classification learning algorithms', *Neural Computation*, 10(7), pp. 1895–1923.

Egressy, B., von Niederhäusern, L., Blanuša, J., Altman, E., Wattenhofer, R. and Atasu, K. (2024) 'Provably powerful graph neural networks for directed multigraphs', *Proceedings of the AAAI Conference on Artificial Intelligence*, 38.

Fernandez, A., Garcia, S., Herrera, F. and Chawla, N.V. (2018) 'SMOTE for learning from imbalanced data: Progress and challenges', *Journal of Artificial Intelligence Research*, 61, pp. 863–905.

Financial Action Task Force (2022) *Money Laundering*. Paris: FATF.

Financial Conduct Authority (2023) *PS7/23: Model Risk Management Principles for Banks*. London: FCA.

Gorishniy, Y., Rubachev, I., Khrulkov, V. and Babenko, A. (2021) 'Revisiting deep learning models for tabular data', *Advances in Neural Information Processing Systems*, 34, pp. 18932–18943.

Jullum, M., Løland, A., Huseby, R.B., Ånonsen, G. and Lorentzen, J. (2020) 'Detecting money laundering transactions with machine learning', *Journal of Money Laundering Control*, 23(1), pp. 173–186.

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q. and Liu, T.Y. (2017) 'LightGBM: A highly efficient gradient boosting decision tree', *Advances in Neural Information Processing Systems*, 30, pp. 3146–3154.

LexisNexis Risk Solutions (2022) *True Cost of Financial Crime Compliance Study*. Atlanta: LexisNexis Risk Solutions.

Lundberg, S.M. and Lee, S.I. (2017) 'A unified approach to interpreting model predictions', *Advances in Neural Information Processing Systems*, 30, pp. 4765–4774.

Lundberg, S.M., Erion, G., Chen, H., DeGrave, A., Prutkin, J.M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N. and Lee, S.I. (2020) 'From local explanations to global understanding with explainable AI for trees', *Nature Machine Intelligence*, 2(1), pp. 56–67.

Oztas, B., Cheng, J. and Heywood, N. (2024) 'Intelligent transaction monitoring for anti-money laundering', *Journal of Money Laundering Control*, 27(3), pp. 459–474.

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A.V. and Gulin, A. (2018) 'CatBoost: Unbiased boosting with categorical features', *Advances in Neural Information Processing Systems*, 31, pp. 6638–6648.

Saito, T. and Rehmsmeier, M. (2015) 'The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets', *PLOS ONE*, 10(3), e0118432.

Savage, D., Zhang, X., Yu, X., Chou, P. and Wang, Q. (2016) 'Detection of money laundering groups using supervised learning in networks', *Expert Systems with Applications*, 72, pp. 344–359.